

ECOSWITCH

Intelligent Classroom Energy Management

Power the classroom. Not the emptiness.

An intelligent classroom energy-management system that detects occupancy, understands inactivity, and automatically manages power zone by zone.

ABOUT THIS FILE

This is a placeholder document. It stands in for the project's full engineering report so that every "Technical PDF" link on the EcoSwitch website resolves to a real file. Replace it with the actual report - no markup changes are required.

01 System overview

- * EcoSwitch is a supervisory layer between a classroom and its electrical loads. It observes presence and activity, decides whether each zone is genuinely in use, and only then acts.
- * The classroom is divided into horizontal sensing zones. Each zone has its own sensing beams, its own occupancy state, its own inactivity timer and its own controlled loads.
- * A single controller (ESP32-S3) samples every sensor input, runs the decision logic, and drives the buzzer, display and relay outputs for each zone independently.
- * The system warns before it acts: occupants receive an audible cue at ten minutes of inactivity, and controlled loads return to standby at fifteen minutes.

02 Design parameters

Sensing zones per classroom	6 potential zones in the design concept (4 beams shown in the reference layout)
Occupancy interpretation	Filtered and validated per zone before any state change
Warning threshold	10 minutes of continued inactivity
Power-control threshold	15 minutes of continued inactivity
Reset condition	Any validated occupancy or activity in the zone
Control granularity	Per zone - occupied zones are never affected by an empty one

03 Scope of this concept

EcoSwitch is a prototype concept. The figures on this page and on the website describe design parameters and simulated interface behaviour; they are not measured savings from an installed system. No component specification, electrical rating or installation detail should be taken from this document - those must be established for the specific classroom by a qualified professional.

SECTION 04

System architecture

Sensors v	mmWave radar modules, IR beam-break pairs, door activity input, PIR cross-check
ESP32-S3 v	the single controller: samples every input, runs the decision logic, drives every output
Occupancy interpretation v	raw signals are filtered and combined into a validated per-zone decision
Zone state engine v	holds occupied / idle / warning / powered-down for every zone
Inactivity timers v	independent timer per zone - reset by activity, advanced only by silence
Warning system v	buzzer and local display announce the ten-minute warning
Power control v	relay stage switches the controlled loads of that zone only
Lights / fans / other controlled loads	powered only while their zone needs them

05 Sensing strategy

Two sensing approaches answer two different questions. EcoSwitch uses both: radar for continuous presence, beam-breaks for deterministic, row-level activity. Neither is universally superior.

A - mmWave presence sensing

- * 24 GHz FMCW radar front end
- * Detects moving and relatively stationary people
- * Useful for continuous occupancy detection
- * Does not require optical line-of-sight
- * Can detect presence without relying only on movement

Watch-outs

- Placement and environmental reflections affect accuracy
- Objects can produce radar reflections

B - IR / laser beam-break

- * Emitter sends a beam to a receiver
- * Beam interruption indicates occupancy / activity
- * Extremely simple and inexpensive
- * Works well for fixed seating and rows

Watch-outs

- Requires line-of-sight and precise alignment
- A beam-break alone cannot determine continued presence once the beam is clear
- Layout changes can disrupt alignment

- Requires calibration

For fixed seating, an IR beam-break can be extremely deterministic. For continuous presence, mmWave is more flexible.

mmWave is more complex than a simple beam-break sensor

SECTIONS 06 - 07

Control logic and interpretation

06 The fifteen-minute logic

0:00	ACTIVE	Occupancy detected in the zone. The timer sits at zero and the controlled loads remain on.
5:00	IDLE	No validated activity has reached the zone. The inactivity timer keeps counting.
10:00	WARNING	The buzzer warns the room. Any validated activity resets the timer immediately.
15:00	POWER CONTROL	The zone's controlled loads are disabled independently of every other zone.

Any validated occupancy or activity resets the timer. The warning exists so that occupants can respond before automation acts; the power-control step only follows sustained, confirmed inactivity.

07 Sensor data interpretation

The sensors never control the power directly. A beam interruption is evidence, not a decision. The signal passes through a defined pipeline before any control action is issued:

RAW SENSOR INPUT	beam state, radar presence, door activity
▼	
FILTERING	debounce, hold time, rejection of single-frame glitches
▼	
OCCUPANCY DECISION	combined evidence becomes a per-zone yes / no
▼	
ZONE STATE	occupied, idle, warning or powered down
▼	
TIMER	each zone counts its own quiet time
▼	
CONTROL ACTION	relay command for that zone's loads, and nothing else

08 Safety and real-world deployment

- * The prototype uses low-voltage loads. It demonstrates behaviour and logic, not an installable electrical design.
- * Real classroom mains switching requires properly rated contactors, protective devices, enclosures and wiring matched to the circuits they serve.
- * The controller does not switch mains directly. The ESP32-S3 drives a control stage; it is never the component that opens or closes a mains circuit.
- * Essential safety circuits - emergency lighting, alarms, exit signage - should not be part of the automated shutdown system at all.

SECTIONS 09 - 10

Hardware, roadmap and next steps

09 Hardware roles

ESP32-S3 CONTROLLER	Reads every sensor input, runs the occupancy and timer logic, drives outputs.
S3KM1110 mmWave sensor PRESENCE SENSING	Compact 24 GHz FMCW radar used for continuous presence detection.
IR emitter BEAM SOURCE	Projects an infrared beam across the room to a paired receiver.
Photodiode / phototransistor BEAM DETECTION	Senses the beam at the far wall; loss of signal is an activity event.
PIR sensor SECONDARY TRIGGER	Low-cost motion cue used as a cross-check, not as the primary decision.
Buzzer LOCAL WARNING	Audible cue at the ten-minute mark, before any power control.
OLED display LOCAL STATUS	Shows zone state, running timer and active power state.
Relay module LOAD SWITCHING	Isolated switching stage between controller and controlled loads.

This list describes component roles and function, not a verified electrical design. Ratings, protection and wiring must be specified for the installation by a qualified professional.

10 Roadmap

V1 Prototype	Sensing, occupancy interpretation, the 10/15-minute logic, buzzer warning and relay control.
V2 Zone intelligence	Door-event context, smarter debounce and hold windows, per-zone calibration profiles.
V3 Energy analytics	Current sensing per zone, logging and reporting so behaviour can be measured.
V4 Campus-wide deployment	Multi-classroom management, occupancy heatmaps and central oversight.

Planned future features: real-time dashboard, energy measurement, historical analytics, occupancy heatmaps, multi-classroom management, Wi-Fi / MQTT integration, Google Sheets or cloud backend, automatic reporting. These are planned directions of work, not shipped features.

11 Replacing this template

- * Add the project's actual engineering report - architecture, schematics, sensing strategy, control logic, hardware design, calibration procedure and test results.

- * Keep the file name pdf.pdf, or update the four "Technical PDF" / "Documentation" links in index.html